Continuous test functions from monomial moments

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Abstract

Second step of the continuous-amplitude milestone (Astra #16, route C), independent of the Laplace-specific constants. On the closed unit cube $K = [0,1]^d$: every continuous η is uniformly approximated by a multivariate polynomial (`exists_mvPolynomial_near`, Stone–Weierstrass for the subalgebra generated by the coordinate functions, whose elements are exactly polynomial functions), and nonnegative weighted functionals $T_N(\eta) = \int_S \eta(\varphi_N x)\, w_N(x)\, dx$ on $S \subseteq K$ with maps φ_N, φ_L into K converge against every continuous test function as soon as they converge on monomials (`tendsto_integral_of_monomials`). Zero `sorry`/`axiom`.

1 The things formalised

```
def closedCube (d :  ) : Set (Fin d →  ) := Set.pi univ fun _ => Icc 0 1
def coordAlgebra (d :  ) : Subalgebra   C(Fin d →  ,  ) :=
  Algebra.adjoin   (Set.range fun i => (ContinuousMap.eval i : C(Fin d →  ,  )))
theorem coordAlgebra_separatesPoints (d) : (coordAlgebra d).SeparatesPoints
theorem mem_coordAlgebra_eval (hg : g   coordAlgebra d) :
      p : MvPolynomial (Fin d)  ,   x, g x = MvPolynomial.eval x p
theorem exists_mvPolynomial_near (d) ( ) (h  : Continuous  ) ( ) (h  : 0 <  ) :
      p : MvPolynomial (Fin d)  ,   x   closedCube d, |  x - MvPolynomial.eval x p| <
theorem tendsto_of_approx (f :   →  ) (a :  ) (h  :     > 0,   g b, Tendsto g atTop (  b)
      (   N in atTop, |f N - g N|    )   |a - b|    ) : Tendsto f atTop (  a)
theorem integrableOn_comp_mul_of_continuous ... : IntegrableOn (fun x => f (  x) * w x) S
theorem integral_eval_mul_eq_sum ... :   x in S, MvPolynomial.eval (  x) p * w x
    =      p.support, MvPolynomial.coeff   p *   x in S, (  i,   x i ^   i) * w x
theorem tendsto_integral_of_monomials (d S hS    L h  h L h m h Lm w wL hw hwL hwnn hwLnn)
    (hmono :     : Fin d →  , Tendsto (fun N =>   x in S, (  i,   N x i ^   i) * w N x)
      atTop (  (  x in S, (  i,  L x i ^   i) * wL x))) ( ) (h  : Continuous  ) :
    Tendsto (fun N =>   x in S,   (  N x) * w N x) atTop (  (  x in S,   (L x) * wL x))
```

2 Mathematics

The constant monomial gives $T_N(1) \to T_L(1) =: M$, hence an eventual mass bound $T_N(1) \le M + 1$. For a polynomial $p = \sum_\gamma c_\gamma u^\gamma$, finite linearity of the integral reduces $T_N(p) \to T_L(p)$ to the monomial hypothesis. For continuous η and $\varepsilon > 0$, choose p with $\sup_K |\eta - p| < \delta := \varepsilon/(M+2)$; positivity of the weights gives $|T_N(\eta) - T_N(p)| \le \delta\, T_N(1) \le \delta(M+1) \le \varepsilon$ eventually and $|T_L(\eta) - T_L(p)| \le \delta M \le \varepsilon$; the $\varepsilon/3$ lemma finishes.

3 Paper-facing meaning

This is the analytic device that upgrades the shifted monomial moments of unit 184 to continuous amplitudes: the limiting functional may be supported on the face $u_J = 0$ (φ_L the projection killing the minimal coordinates) while the finite-N functionals use $\varphi_N = \mathrm{id}$.

4 Lean notes

- `∞` is a token (the `ENNReal` top), so `∞` is not an identifier; use `L`.
- Stone–Weierstrass: `ContinuousMap.exists_mem_subalgebra_near_continuous_of_isCompact_of_separatesPoints` with the adjoin of the coordinate evaluations; `Algebra.adjoin_induction` via `induction hg using ... with | mem | algebraMap | add | mul` produces the polynomial.
- `MvPolynomial.continuous_eval` takes the polynomial explicitly and lives in `Mathlib.Topology.Algebra.MvPolynomial`, which must be imported explicitly.
- Sums over `p.support` index by `Fin d →` ; the monomial hypothesis quantifies over `Fin d →` , so instantiate it at `fun i =>  i` and annotate the binder.